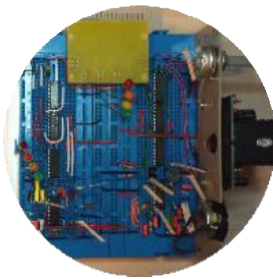


Lab 4-7 – Project lab: Audio circuit with selectable input signals



Week 10-13 will be dedicated to the project lab. During these 4 weeks you will design, build, measure and document an audio circuit with selectable input signals. Elements in the circuit will be a microphone input with impedance bridging, amplifier, filter, channel selector and an Arduino based audio input of your choice.

It is important to understand that part of the process is to find the information and to build a circuit without being told exactly what circuit to build. However, do not panic, there are many sources of information ranging from your course book, through data sheets and friends to the assistants. Always remember: **Do not panic!** This is only a lab.

Read this text twice before asking the assistant.

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General description

The project consists of four compulsory sub-projects: microphone input, filter, additional input and channel selector. The microphone must be impedance bridged to the load and pre-amplified before filtering.

The second input signal will be something you select and must involve the Arduino. Finally, selection of inputs is to be controlled with a summing amplifier circuit. A schematic of the entire circuit is given in Figure 1.

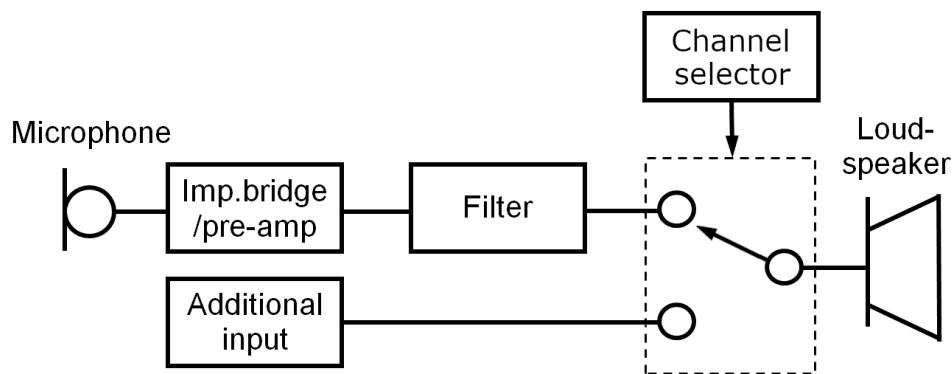


Figure 1: Schematic of the audio amplifier circuit.

We recommend that you try to complete approximately one sub-project per week. Each step should be simulated, built, tested, and documented before it is integrated into the total project. To fit everything on the breadboard at once, it is a good idea to build compact circuit parts. First build and test, then tidy up a bit. Then test again. (And remember that the op-amps contain four amplifiers!)

Documentation will consist of measured frequency response (using function generator and oscilloscope) in the form of **Bode plots**, as well as **Multisim schematics** and **simulation results**. To easily compare simulations and measurements, Multisim simulation data can be exported to for example Excel or other file formats. You are free to use any plotting solution you like but make sure to follow the guidelines for plotting scientific data.

At the end of each main section, there is a list of expected documentation for the task. The documentation of the project will be needed in the lab evaluation at the end of the semester. There is also a check list at the end of this document. Check it at least twice. **Remember that a Bode plot contains both gain and phase.**

Reference voltage (DC offset)

Use a voltage divider to create a DC offset as reference signal (as seen in exercise 4: inverting amplifier in lab 3). Usually ground is used as reference, but as we use single sided supply for the op-amps, they cannot give out negative voltages, and hence we need a positive offset for sinusoidal (or other AC) signals.

With the setup shown in figure 2 we get a V_{ref} of approx. 1V. You will need V_{ref} for the microphone circuit (section 1) and the filter (section 2). It is up to you whether you build one for each module or use the same one for all of them.

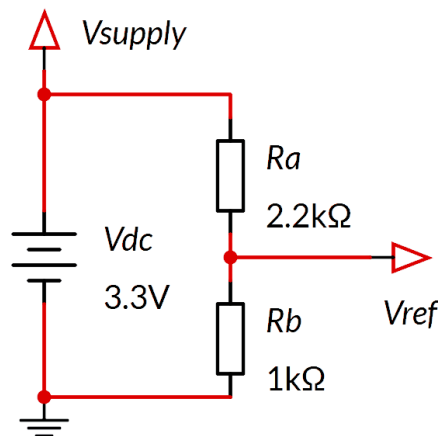


Figure 2: Creating a reference voltage with two resistors. If V_{ref} is noisy, it may help placing a capacitor in parallel with the bottom resistor, to remove high frequency signals.

Equipment and components

The following equipment will be available in the cupboards or on the central components table, and must be returned after each lab session:

- Handheld multimeter
- Oscilloscope with function generator
- Loudspeaker
- Wire tools
- USB cable and AC adapter for Arduino Nano

The following equipment will be selected by each individual group, and is to be kept on separate shelves after each lab session:

- Breadboard
- Arduino Nano 33 IoT or Arduino Nano 33 BLE Sense Rev2
- Capacitor microphone element ($Z = 2.2 \text{ k}\Omega$, frequency range 20-16000 Hz)
- Standard circuit elements (R, C, diodes etc. as required)
- LM324 operational amplifiers
- Pushbutton switches

Project assignments

1 Microphone input

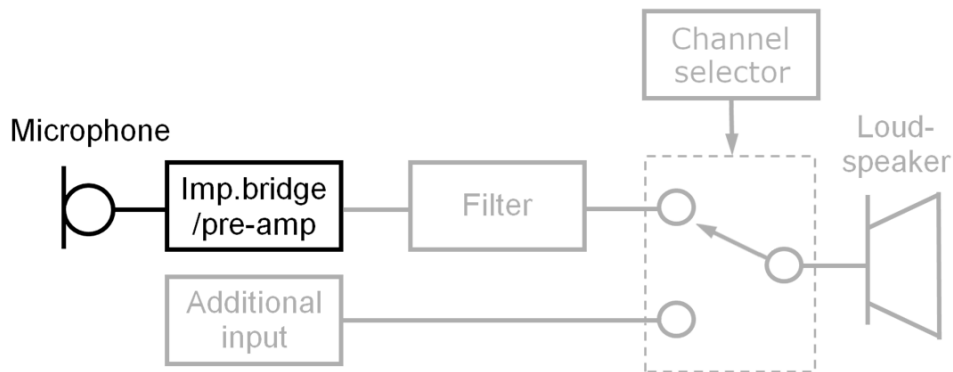


Figure 3: Part one: The microphone input

Due to the impedance of the microphone, it requires so-called bridging to allow for maximum signal transfer. You should also amplify the signal to an appropriate level, using a simple op-amp (LM324). We suggest 1-2 V peak-to-peak output from this stage.

To help get started, an input circuit is suggested in Figure 4. The microphone input is buffered by a unity gain non-inverting amplifier and amplified in an inverting amplifier. The microphone input is grounded through a large resistor (10 M Ω) to avoid a floating voltage level. The buffer output is then AC coupled to the inverting amplifier.

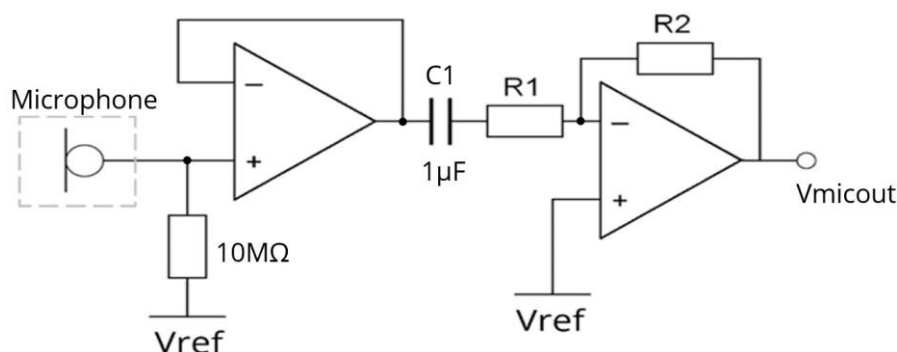


Figure 4: Sketch of the microphone and buffer circuit, AC coupled to an inverting amplifier.

R1 and C1 form a high-pass filter, so to avoid losing the lower frequency range, R1 should be 10 k Ω or more. This will secure a cut-off frequency below

16 Hz, and that is good enough, as the audible frequency span is defined as 20 – 20 000 Hz.

Then R2 must be selected to get the wanted amplifier output voltage.

The capacitor microphone needs a supply voltage to work. The “ideal microphone” symbol in Figure 4 should on the breadboard look like Figure 5 a) and in the simulation software like Figure 5 b).

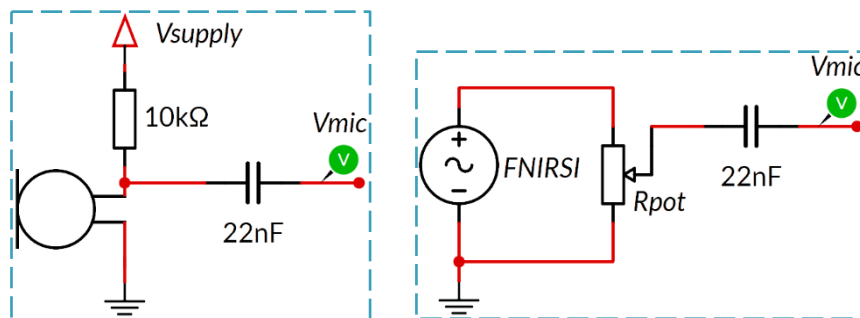


Figure 5: a) Microphone with power supply. b) “Dummy microphone” to be used in Multisim simulation. Adjust the function generator (V_{mic}) value so that the output from this circuit is similar to the output from the real microphone circuit in a).

To find the appropriate voltage to use for V_{mic} , build the circuit in Figure 5 a) on the breadboard, connect the output to an oscilloscope and measure while whistling or singing. (This is also an easy test to check if the microphone is working.)

Documentation

- A short explanation (a few lines) of the physical principles of your microphone.
- Multisim circuit schematics
- Simulated (Multisim) frequency response (Bode plot)
- Measured frequency response (Bode plot). Use the “dummy microphone” setup as input signal for these measurements.

Use 10 Hz – 100 kHz for the Bode plots. **Remember that the Bode plot contains both gain and phase.**

At last connect the real microphone to the circuit. Check that the output signal is of suitable magnitude (1-2 V p-p) or change the amplification until it is.

Relevant literature can be found in chapter “Operational Amplifiers” (Storey), datasheet for LM324 (Blackboard) and on e.g. Wikipedia (“microphone”, “Impedance bridging”, etc).

2 Filter

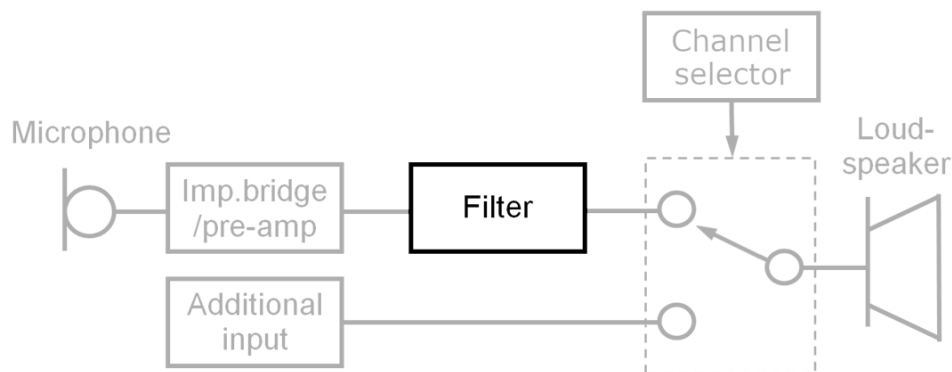


Figure 6: Part two: Filter

A filter must be added after the impedance bridging/ pre-amplifier step, to reduce noise, as well as controlling the level of bass and treble. To control bass and treble individually, we suggest one high-pass and one low-pass filter in series, for example Sallen-Key¹ filters. The filter must be an active filter, of second order or higher. You may use the LM324 op-amp in the filter.

Make sure the filter circuit works correctly before connecting it to the rest of the project.

Note: It is up to you to select frequency area for the filter, but it is nice to make it actually filter something, so using 20 Hz and 20 000 Hz may be a bit boring, as we cannot hear it do anything. Be creative!

¹ https://en.wikipedia.org/wiki/Sallen-Key_topology

Documentation

- Selected values of corner (cut-off) frequencies
- Multisim circuit schematics
- Simulated (Multisim) Bode plot of the stand-alone filter circuit
- Measured Bode plot of the stand-alone filter circuit. Use the function generator as input for measuring this Bode plot.

Use 10 Hz – 100 kHz for the Bode plots. **Remember that the Bode plot contains both gain and phase.**

3 Additional input

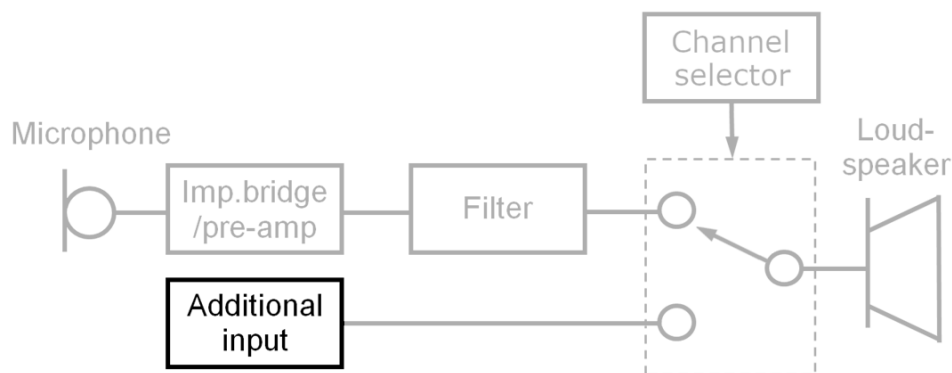


Figure 7: Part four: Additional input

Select one of the suggested Arduino based inputs or choose your own:

- Electric piano
- Single frequency tuned by photoresistor
- 4-bit sine wave signal using R-2R

Important: Avoid using the `tone()` function in this task, it tends to freeze/crash the Arduino Nano.

Documentation

Depending on the choice: Description, Arduino code, Multisim schematics, simulations, and measured results.

4 Channel selector

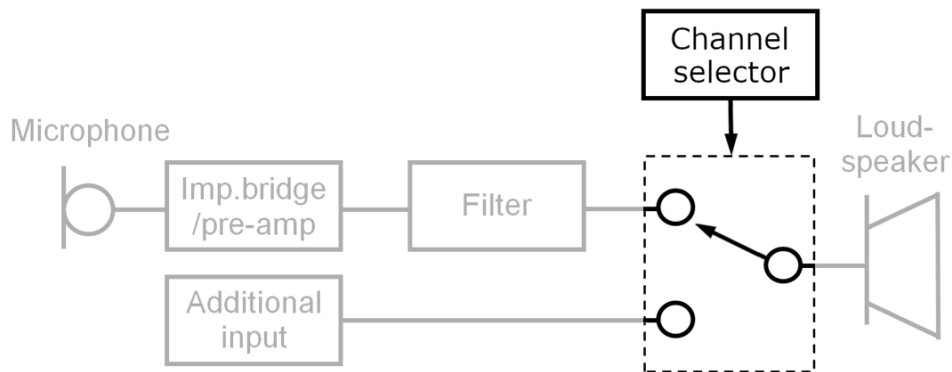


Figure 8: Part four: Channel selector

Use a non-inverting summing amplifier in combination with switch buttons to select one of the two or more possible inputs (microphone and additional input). A simple sketch of the control circuit is given in Figure 9.

You can add more inputs by adding a switch and a resistor for each input (E.g. if you want to be able to listen to your microphone with and without filter)

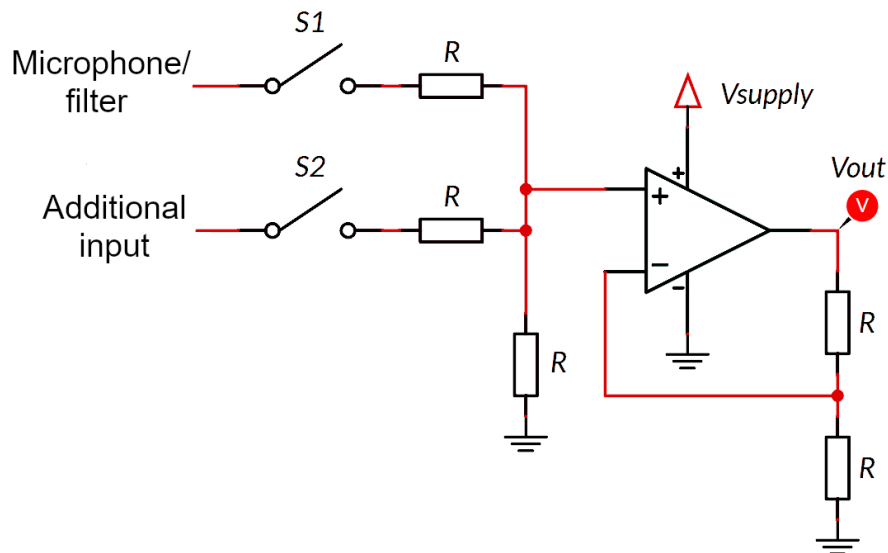


Figure 9: Channel selection with a non-inverting summing amplifier (adder). If the resistors are equal, V_{out} will be equal to the input signal when one switch is closed. One can add several inputs to this circuit by adding a switch and a resistor for each new input.

Documentation

- A short description of how the circuit works

- Simulation of the circuit with dummy microphone input and a second function generator signal. *What happens if you close both switches at the same time? Can the same be observed in your circuit?*

5 Complete project

Connect all the sub-projects together and connect a loudspeaker to the adder output to test the complete build.

6 Extra tasks

If you would like to do something more, build some extra inputs or try out something beyond the tasks asked here, please feel free to do so! This can be anything also non sound inputs. Ask us for available parts and we try to help as good as we can!

Appendix: Documentation check list

Remember that Bode plots contain both gain and phase.

1. Microphone input

- ☐ Physical principle of selected microphone
- ☐ Multisim circuit schematics
- ☐ Bode plot for simulated frequency response (Multisim)
- ☐ Bode plot for measured frequency response (function generator source)

2. Filter

- ☐ Selected values of cut-off frequencies
- ☐ Multisim circuit schematics
- ☐ Bode plot for simulated frequency response (Multisim)
- ☐ Bode plot for measured frequency response (function generator source)

3. Additional input

- ☐ Depending on the type of assignment: Description, Multisim schematics, simulations, and measured results.
- ☐ Arduino Code

4. Channel selector

- ☐ Short function description
- ☐ Multisim circuit schematics.
- ☐ Simulation results. Test what happens when only one or both inputs are active.